

# Contest Mindset and Diagnostic

## Entrance Diagnostic · Form A

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|-------------|--|-------------|--|
| <b>Name</b> |  | <b>Date</b> |  |
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**60 minutes. No calculator.** 25 questions in three parts. Questions 1–10 are worth 2 points, 11–20 are worth 5 points and 21–25 are worth 6 points (100 points in total). For questions 1–20, circle one answer and write its letter on the line or in the box. For questions 21–25, write your answer on the answer line. Where you see a method box, show your diagram, table or equation: it can earn points even if your final answer is wrong. Diagrams in this paper are drawn to scale. Skip a question if you are stuck and come back to it.

### Part 1 — Foundations. Questions 1–10, 2 points each.

1. What is  $\frac{3}{4}$  of 36?  
 (A) 9                      (B) 12                      (C) 24                      (D) 27                      (E) 48

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2. What is 15% of 80?  
 (A) 8                      (B) 12                      (C) 15                      (D) 16                      (E) 65

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3. If  $4n + 7 = 43$ , what is the value of  $n$ ?  
 (A) 4                      (B) 7                      (C) 9                      (D) 12.5                      (E) 36

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4. The pattern 5, 12, 19, 26, ... continues by adding 7 each time. What is the 10th number in the pattern?  
 (A) 50                      (B) 61                      (C) 65                      (D) 68                      (E) 75

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5. A rectangle is 12 cm long and 7 cm wide. What is its perimeter?  
(A) 19 cm      (B) 26 cm      (C) 38 cm      (D) 84 cm      (E) 168 cm
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6. Two angles of a triangle measure  $48^\circ$  and  $67^\circ$ . What is the measure of the third angle?  
(A)  $45^\circ$       (B)  $55^\circ$       (C)  $65^\circ$       (D)  $115^\circ$       (E)  $295^\circ$
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7. What is the mean (average) of 8, 11, 12, 15 and 24?  
(A) 11      (B) 12      (C) 13      (D) 14      (E) 70
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8. Which of the following is a prime number?  
(A) 59      (B) 39      (C) 51      (D) 57      (E) 91
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9. What is the lowest common multiple of 6 and 8?  
(A) 2      (B) 12      (C) 14      (D) 16      (E) 24
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10. Ana, Ben and Cy each own exactly one pet: a cat, a dog or a fish, all different. Ben does not own the dog. Cy owns the cat. Who owns the fish?  
(A) Ben      (B) Ana      (C) Cy      (D) Ana or Ben      (E) It cannot be determined
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**Part 2 — Problem solving. Questions 11-20, 5 points each.**

11. A jacket costs \$80. It is on sale for 25% off. Then 13% tax is added to the sale price. What is the final cost?

(A) \$60.00      (B) \$67.80      (C) \$70.40      (D) \$80.00      (E) \$90.40

Method box — a correct diagram, table or equation earns points even if the answer is wrong.

**Answer:** \_\_\_\_\_

12. The ratio of red marbles to blue marbles in a bag is 3 : 5. There are 48 marbles. How many more blue marbles than red marbles are there?

(A) 6      (B) 12      (C) 18      (D) 30      (E) 48

Method box — a correct diagram, table or equation earns points even if the answer is wrong.

**Answer:** \_\_\_\_\_

13. Mia ate  $\frac{1}{3}$  of a pizza. Leo then ate  $\frac{1}{4}$  of what was left. What fraction of the whole pizza remains?

(A)  $\frac{1}{6}$       (B)  $\frac{5}{12}$       (C)  $\frac{1}{2}$       (D)  $\frac{7}{12}$       (E)  $\frac{2}{3}$

**Answer:** \_\_\_\_\_

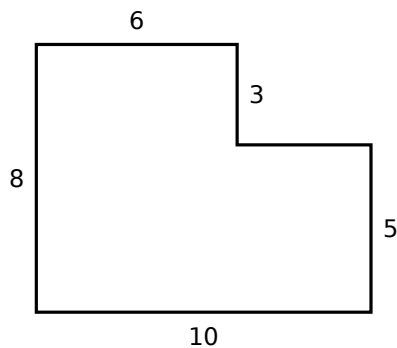
14. Squares are built in a row from toothpicks: 1 square uses 4 toothpicks, 2 squares use 7, and 3 squares use 10. How many toothpicks are needed for 20 squares in a row?

(A) 60                      (B) 61                      (C) 63                      (D) 80                      (E) 81

Method box — a correct diagram, table or equation earns points even if the answer is wrong.

**Answer:** \_\_\_\_\_

15. The figure is made of rectangles and all corners are right angles. Some side lengths, in cm, are shown. What is the area of the figure, in  $\text{cm}^2$ ?

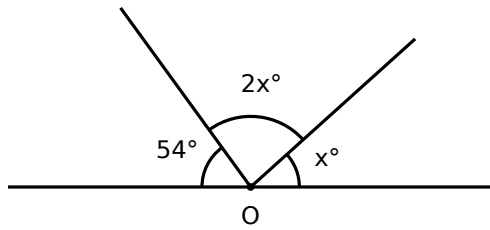


(A) 56                      (B) 62                      (C) 68                      (D) 74                      (E) 80

Method box — a correct diagram, table or equation earns points even if the answer is wrong.

**Answer:** \_\_\_\_\_

16. In the diagram, O lies on a straight line. Find the value of  $x$ .



- (A) 21                      (B) 42                      (C) 63                      (D) 84                      (E) 126

**Answer:** \_\_\_\_\_

17. A rectangular prism has a volume of  $360 \text{ cm}^3$ . Its base is 8 cm by 9 cm. What is its height?

- (A) 3 cm                      (B) 4 cm                      (C) 5 cm                      (D) 20 cm                      (E) 45 cm

**Answer:** \_\_\_\_\_

18. A bag has 4 red marbles, 5 blue marbles and some green marbles. The probability of picking a blue marble is  $\frac{1}{3}$ . How many green marbles are in the bag?

- (A) 2                      (B) 3                      (C) 5                      (D) 6                      (E) 15

Method box — a correct diagram, table or equation earns points even if the answer is wrong.

**Answer:** \_\_\_\_\_

19. How many whole numbers from 1 to 100 are multiples of 3 or multiples of 5 (or both)?

(A) 33                      (B) 41                      (C) 47                      (D) 53                      (E) 58

Method box — a correct diagram, table or equation earns points even if the answer is wrong.

**Answer:** \_\_\_\_\_

20. Four friends finish a race with no ties. Ari finishes before Bo. Bo finishes before Dee. Cal finishes immediately after Dee. Who finishes third?

(A) Ari                      (B) Bo                      (C) Cal                      (D) Dee                      (E) It cannot be determined

**Answer:** \_\_\_\_\_

### ★ Challenge

Part 3 — Challenge. Questions 21–25, 6 points each. Write your answer on the line.

21. In a club, the ratio of boys to girls is 3 : 4. After 6 more boys join (and nobody leaves), the numbers of boys and girls are equal. How many members does the club have now?

Method box — a correct diagram, table or equation earns points even if the answer is wrong.

**Answer:** \_\_\_\_\_

22. The sum of five consecutive even numbers is 170. What is the largest of the five numbers?

Method box — a correct diagram, table or equation earns points even if the answer is wrong.

**Answer:** \_\_\_\_\_

23. A  $4 \times 4 \times 4$  cube is built from 64 small cubes. The outside of the large cube is painted. How many small cubes have paint on exactly two faces?

Method box — a correct diagram, table or equation earns points even if the answer is wrong.

**Answer:** \_\_\_\_\_

24. Two fair six-sided dice are rolled. What is the probability that the two numbers rolled add to 8?

Method box — a correct diagram, table or equation earns points even if the answer is wrong.

**Answer:** \_\_\_\_\_

25. How many three-digit whole numbers have digits that add to 5?

Method box — a correct diagram, table or equation earns points even if the answer is wrong.

**Answer:** \_\_\_\_\_